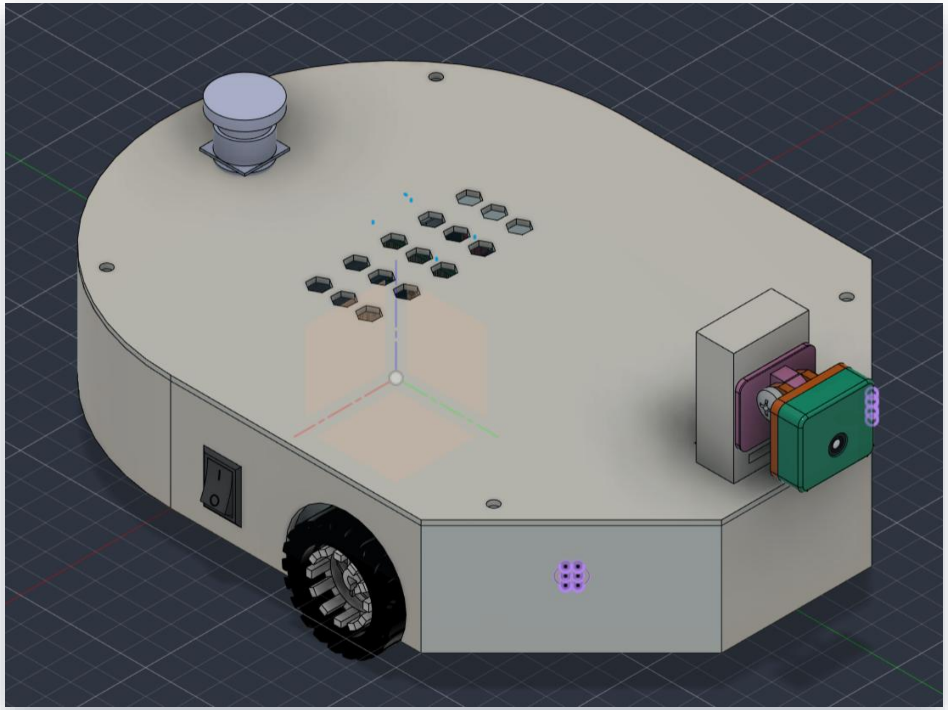
ROBO RUMBLE · ROBO GRAND PRIX CATEGORY

APEX-ONE

AUTONOMOUS RACING SYSTEM



TEAM APEX

Tshwane University of Technology

“Engineered for high-speed precision, local viability, and robust autonomous navigation.”

TEAM APEX

THE CHALLENGE: SPEED VS. PROCESSING BOTTLENECKS

The Arena Constraint

The Robo Grand Prix requires vehicles to navigate a complex track at high speeds while avoiding unexpected obstacles.

The Processing Problem

Standard microcontrollers cannot handle heavy computer-vision algorithms, while microcomputers like Raspberry Pi suffer timing delays when directly controlling high-speed motors.

The Supply Chain Challenge

Relying purely on imported robotics parts results in massive delays and inflated costs, stifling rapid prototyping in African markets.

THE APEX-ONE SOLUTION: DUAL-CORE ARCHITECTURE

Dual-Core Processing Architecture

Allocates real-time motor control and low-level sensor polling to an STM32 microcontroller while deploying a Raspberry Pi Zero 2 W for high-level computer vision.

Standalone Edge Vision

A lightweight, custom-built OpenCV pipeline (no bulky middleware) maximises the Pi Zero's 512MB RAM for ultra-fast frame processing and latency-free path planning.

Hardware Reliability

Industrial-grade components — a High-Endurance microSD card, TB6612FNG dual motor driver, and INA219 current sensor — protect the system under load.

Rapid Buildability

A modular 3D-printed PLA chassis, optimised for standard slicer profiles and assembled with non-conductive M3 nylon standoffs.

SYSTEM DATA FLOW

RASPBERRY PI ZERO 2 W

Standalone OpenCV · Path Planning

I2C / UART ↔

STM32 BLUE PILL

Real-Time PID · Sensor Polling

Real-Time I/O ↔

MOTORS & SENSORS

Drive Output · ToF / IR Feedback

RAPID FABRICATION & STRATEGIC SOURCING

Strategic Sourcing

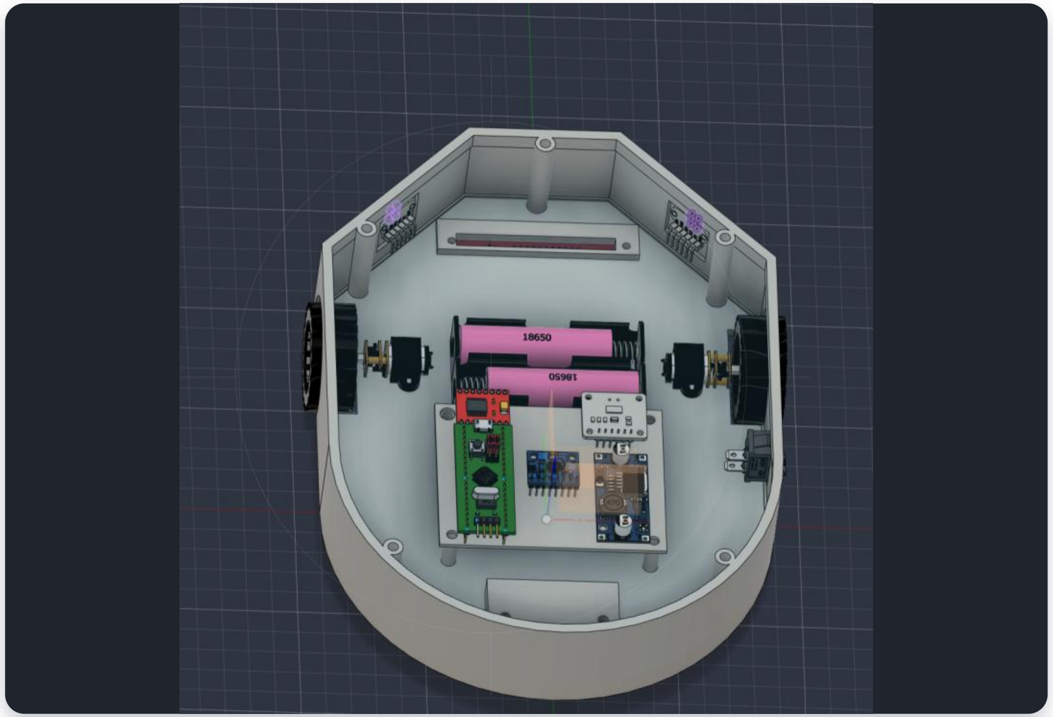
95% of components — sensors, logic, power — are sourced locally (DIYElectronics, Communica, Bob Shop) for 2-day delivery. Only the specialised, high-RPM N20 gear motors were imported via AliExpress to optimise the speed-to-cost ratio.

Modular Mechanical Design

The chassis tub and lid are 100% 3D-printed in standard PLA (~250g).

Scalable Assembly

Elevated perfboard mounting using non-conductive M3 nylon standoffs ensures a clean, easily repairable electrical deck for trackside maintenance.



KEY SUPPLIERS — LOGO PLACEHOLDERS

DIYElectronics | Communica | AliExpress

FINANCIAL DOCUMENTATION (BILL OF MATERIALS)

TOTAL IMPLEMENTATION COST

R 3,545.39

(Includes Local VAT; Excludes Shipping)

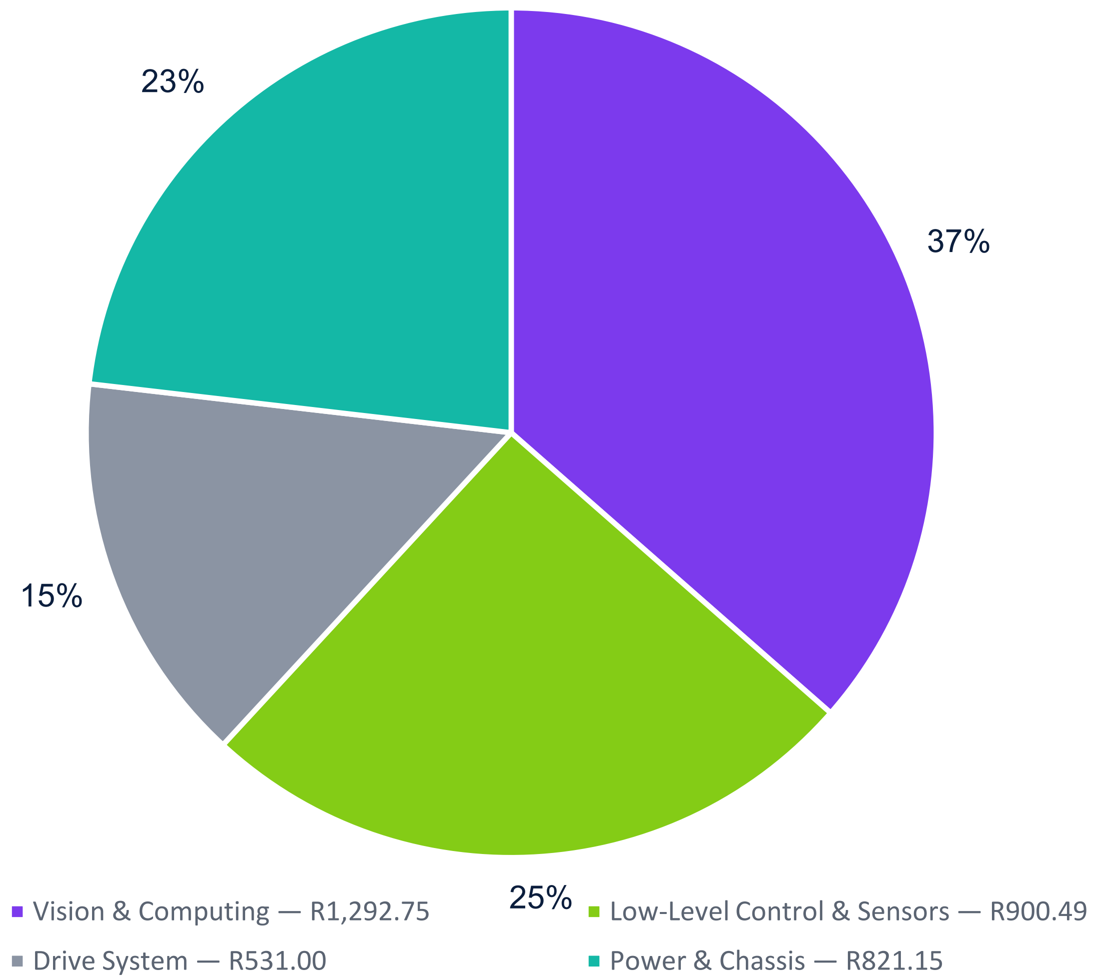
Verdict: A highly competitive price-to-performance ratio achieved by avoiding expensive pre-built robotic platforms in favor of custom integration.

Item No.	Component Name	Quantity	Cost per Unit (ZAR)	Total Item Cost (ZAR)	Supplier Nam
	⋮ rows omitted ⋮				
17	INA219 I2C High-Side Current/Power Sensor	1	R 30.00	R 30.00	Communica
18	Double-Sided Prototype Perfboard (Pack)	1	R 59.00	R 59.00	DIYElectronics
19	Lead-Free Rosin Core Solder Wire Spool	1	R 179.00	R 179.00	DIYElectronics
20	PLA 3D Printing Filament (~250g Allocation)	1	R 199.00	R 199.00	DIYElectronics
21	2-Slot 18650 Battery Holder	1	R 10.01	R 10.01	Communica
			TOTAL PROJECT IMPLEMENTATION COST	R 3,545.39	

All amounts in South African Rand (ZAR) · Generated 6 September 2026

Official Bill of Materials — final total highlighted in red

COST BREAKDOWN BY SUBSYSTEM



MANDATORY SAFETY & ELECTRICAL RIGOR

Critical Safety (E-Stop)

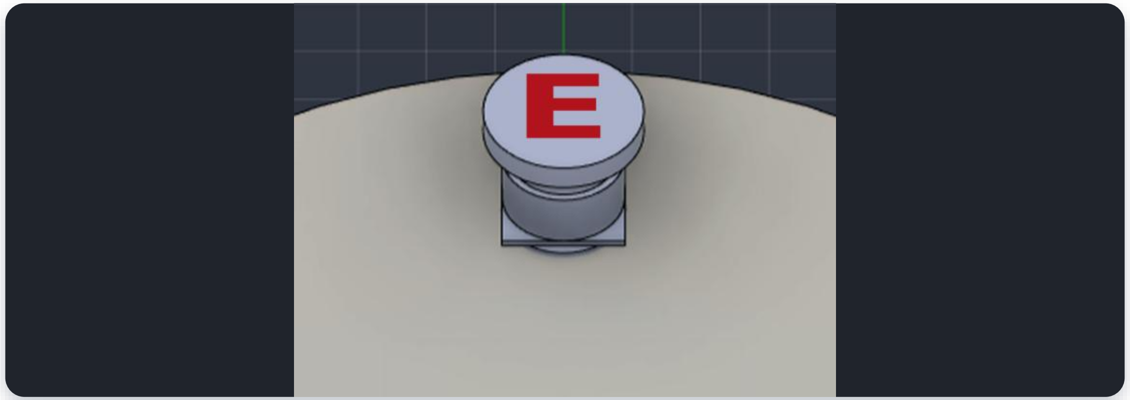
A physical, easily accessible panel-mount Emergency Stop button is hardwired into the main battery rail to instantly sever power to all drive systems in case of software failure.

Power Regulation

7.4V nominal 18650 array is stepped down by LM2596 buck converters to safely provide clean 5V logic to the microcontrollers.

System Telemetry

An INA219 I2C current/power sensor monitors power draw in real time, preventing motor stall currents from frying the motherboard.



SAFETY CUTOFF PATH

18650 BATTERY → E-STOP → MOTOR DRIVER → DRIVE MOTORS

TEAM APEX

The People Behind Apex-One

Mathurin Nimrod Moundzika-Kibamba

Project Lead / Hardware

- Mechanical CAD design, electronic circuit layout, power distribution, and BOM validation.
- Simulation designer and project overseer

Co-lead on physical fabrication and hardware assembly.

Ayanda Delight Tshabangu

Project Lead / Software

- Embedded firmware and low-level hardware driver development (C++/STM32).
- High-level standalone OpenCV computer-vision pipeline and path-planning algorithms.

Co-lead on physical fabrication and hardware assembly.